Tyson **Thompson**

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Professional Experience

Kinaxis

Software Engineer Intern - Data Ingestion *Jan. 2025 - Dec. 2025*

- Designed a data pipeline for Maestro, converting over 100+ GB/day of raw data into clean, structured datasets used for supply chain planning.
- Eliminated over 2 hours of manual deployments per engineer daily by automating deployments of Kubernetes pods in GCP and Azure through CI/CD pipelines using GitHub Actions.
- Reduced transfer times by 90% to GCP, AWS and Azure by utilizing Java multi-threading API to extract data from SAP files concurrently.
- Built a Tkinter GUI for the Data Extractor for over 2000 enterprise customers, replacing JSON configs and speeding up configuration changes.
- Optimized PostgreSQL database queries speed by 67% by introducing a lookup Redis cache and migrating schema to use indexes.
- Developed an integration testing framework for cloud interactions using Mockito, LocalStack, and WireMock.
- Scaled a Kafka- and Spark-powered File Service to 1,000+ auto-scaling Kubernetes pods, orchestrating cross-cloud data syncs from pre-existing YAML configs for multiple engineering teams.

Recollective

Software Engineer Intern - Full Stack *Sep. 2023 - Dec. 2023*

- Developed new interactive features for the qualitative research platform using JavaScript, Vue.js, JSP, and Java, helping clients with more purpose-built ways to gather richer and structured participant data.
- Engineered an LLM-powered research assistant using OpenAI API, designing the prompt and context pipeline that turns study data into accurate, personalized answers, shipped to over a million end-users.
- Built the Sort & Rank interactive drag-and-drop module, enabling researchers to a real-time interface format for ranking and prioritizing items which boosted user time to completion and response completion by over 30% from using the old survey module.
- Increased Google Lighthouse accessibility score by 40 points and achieved WCAG compliance by making components fully keyboard-navigable and screen-reader compatible and integrating an accessibility menu.

Cority Software

Software Engineer Intern - Security *May 2023 - Aug. 2023*

- Engineered an LLM-powered tool that autofills new RFP questionnaires using answers retrieved from a database of previous questionnaires to cut form completion time from 3 days to 10 minutes for hundreds of questions.
- Built a Python script that automatically discovers and aggregates vulnerability assessment reports across multiple teams into a database, eliminating hundreds of hours of manual tracking.
- Resolved 200+ security issues before release by reviewing code and implemented fixes for critical and high-risk vulnerabilities such as SQL injections identified by Snyk.

Cority Software

Software Engineer Intern - IT *May 2022 - Aug. 2022*

- Developed and deployed standardized system images and automated software configuration across diverse hardware platforms to streamline IT operations.
- Developed an automated QA tool in Python that verifies installed software, drivers, and settings on newly imaged machines, replacing manual checklists across multiple hardware models.
- Expanded automated test coverage for the QA tool by adding 20+ test cases, raising coverage from less than 20% to over 90% on SonarQube.

Projects

Computer AI Agent

Python, OpenCV, ChatGPT

- Built an autonomous desktop agent that completes tasks from natural-language instructions using a perceive, reason, act loop on screen state.
- Developed a vision pipeline in OpenCV that segments screenshots and detects UI elements such as buttons, text fields and menus via Template Matching, giving the agent structured context about the screen.
- Integrated LLM function calling to interpret screen context and choose the next action, translating model output into PyAutoGUI inputs.

Education

Carleton University | Bachelor's in Computer Science | 3.8 Major GPA

Sep. 2021 - Dec. 2026

• Relevant Coursework: Systems Programming, Data Structures & Algorithms, Web Applications, Software Engineering, Cryptography and Authentication, Operating Systems, Intro to AI, Algorithms for Data Mining, Web, and Social Networks, Databases, UX Engineering.

Skills

- Languages: Python, JavaScript, TypeScript, Java, Ruby, SQL, Go, C++, HTML, CSS
- Frameworks: React, Node.js, Next.js, Express, Spring Boot, React Native, Flask, Angular, Vue.js, Ruby on Rails
- Tools: Git, Docker, AWS, Kubernetes, GCP, Azure, PostgreSQL, MongoDB, Redis, Kafka, Firebase, Vercel, Stripe
- Concepts: API, Databases, Microservices, Containers, Automated Software Testing, Serverless Functions